

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	DR BEANS
<i>Company:</i>	UoN

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐
	If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.

Project Background Information

In the Mixed Reality Lab (MRL) we have a communal coffee machine which lab members use throughout the day to make coffee. However, like many communal things sometimes the communal responsibility for keeping an eye on how many bags of coffee beans we have left and ordering some more before we run out falls short, and we have supply shortages.

This project's objective is to design and implement an "intelligent", automated system to manage the coffee bean inventory (and maybe more). The system will need to monitor stock levels, forecast consumption, initiate procurement actions and potentially assign blame, all in order to maintain a consistent bean supply.

The project is inspired by a variety of previous coffee related works, including the first webcam (https://en.wikipedia.org/wiki/Trojan_Room_coffee_pot), Anthropic's slightly sinister attempt to have Claude run a small beverage shop (<https://www.anthropic.com/research/project-vend-1>) and Chris Speed's conversational coffee machine (<https://dl.acm.org/doi/abs/10.1145/3025453.3025878>).

¹ An overview of different research areas in the school of computer science can be found [here](#).

The project will need to perform three primary tasks:

- Bean counting. It turns out that bean counting isn't as simple as you might think it is. You'll need to build a system to keep track of the inventory of beans. This could be computer vision based, or some other sensor-based solution such as tracking weight. You'll need to understand how users currently use beans and try to integrate your solution into that.
- Consumption forecasting. You'll need to log stock levels over time to create a historical dataset of coffee usage, then forecast future consumption rates and calculate an estimated data for inventory depletion. The model will need to take into account seasonal changes such as teaching and holidays.
- Coffee conversation. Users should be able to converse with the system in order to make small talk while they are having coffee, or to find out if the beans are running low soon. Maybe the system has an agenda – does it try to optimise coffee consumption in a way that benefits itself?

Note this is not an AI project in and of itself, but it probably involves the application of AI as part of the solution.

Requirements

The group must provide:

- A component that can monitor coffee bean stock and forecast when it run out. A variety of sensing technologies (e.g. Grove PI) are available.
- A conversational component that can engage with users, including asking them to perform tasks such as ordering more beans.
- A stretch goal might include extending the system to cover milk purchases.

These requirements are flexible and can be adapted to your interests/proposed solution.

Skills

Highly desirable:

- Interest in HCI and understanding users
- Interest in building sensor-based systems/ubiquitous computing/IoT

Desirable:

Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.